

DFT-Validated Evidence of Ferromagnetic Mislabeling Among Dzyaloshinskii–Moriya-Candidate Materials in the Materials Project Database

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High-throughput databases such as the Materials Project (MP) are widely used as the first filter in computational searches for Dzyaloshinskii–Moriya-interaction (DMI) hosts for chiral spin-texture and skyrmion-based non-volatile memory. These searches typically trust the database’s recorded magnetic ordering, even though MP’s automated relaxation workflow initializes every calculation from a single ferromagnetic (FM) moment guess and rarely explores antiferromagnetic (AFM) alternatives. We report a direct density-functional-theory (DFT) test of this assumption on a structural family – noncentrosymmetric, heavy-spin-orbit-coupling-element, net-moment compounds – selected because it is exactly the population that DMI-candidate screening pipelines draw from. Using the same energy-mapping protocol as MP’s own workflow, we enumerate and converge FM and symmetry-distinct AFM orderings for 24 candidates drawn from a machine-learning-filtered shortlist built on a seven-source, 64,635-record harmonized materials database. Seventeen of the 24 MP “ground-truth” FM labels are overturned by a lower-energy AFM state (margins of 0.31–24.68 meV/atom), four are confirmed FM, and three are degenerate to within electronic-convergence precision. This DFT-level result independently corroborates, in a materially different and more consequential population, the large-N statistical finding of systematic FM bias in MP recently reported from MAGNDATA-trained ML classifiers [Fahmy, arXiv:2509.05909]. For one confirmed-FM candidate, $\text{Li}_4\text{Fe}_2\text{TeWO}_{12}$ (mp-754090), we go one step further and extract both the isotropic exchange constant ($J = 12.06$ meV, from collinear FM/AFM energy mapping) and the DMI constant ($D = 0.19$ meV, from a constrained noncollinear+spin-orbit canting-angle probe) directly from DFT, and use them, unmodified, as inputs to finite-temperature LAMMPS-SPIN dynamics. The resulting magnetization decays monotonically with a knee between 150 and 200 K, a physically sensible signal despite a known domain-cancellation artifact that suppresses its absolute magnitude. Taken together, these results argue that MP’s ordering label should be treated as a screening prior rather than ground truth for exactly the structural family that DMI-material discovery pipelines target, and they demonstrate what a full, DFT-grounded verification of one candidate costs and returns.

I. INTRODUCTION

Materials that host a stable Dzyaloshinskii–Moriya interaction (DMI) [1, 2] are of active interest for chiral spin-texture and skyrmion-based non-volatile memory, because the antisymmetric exchange term the DMI contributes can stabilize spin spirals and skyrmion lattices at zero field [3, 4]. Searching for such materials computationally begins, almost universally, with a curated high-throughput database – most commonly the Materials Project (MP) [5] – used to filter tens of thousands of candidates down to a tractable shortlist before committing expensive first-principles or simulation resources. That filtering step depends on the database’s recorded magnetic ordering (ferromagnetic, antiferromagnetic, ferrimagnetic, or non-magnetic) being correct, or at least an unbiased estimate.

It is well documented that automated high-throughput DFT workflows are prone to a specific failure mode here: relaxing a structure from a single initial magnetic-moment guess – almost always ferromagnetic – converges to whichever local minimum that guess is closest to, without necessarily finding the true magnetic ground state [6]. Materials Project’s own magnetic-ordering methodology paper is explicit that a full energy-mapping analysis, enumerating symmetry-distinct antiferromagnetic configura-

tions and comparing converged energies, is required to determine the ground state reliably [6]; the bulk of MP’s own entries, generated at scale, do not receive this treatment.

Recent independent evidence, obtained through a very different route, points at the same problem. Fahmy [7] trained propagation-vector and magnetic-order classifiers on the MAGNDATA database of experimentally solved magnetic structures [8, 9] and applied them to the Materials Project database, finding a systematic disagreement consistent with a ferromagnetic bias affecting more than 7,843 MP-labeled compounds. That is a large-N, purely statistical detection: the classifier disagrees with the label, but no first-principles calculation is performed on any individual compound to confirm which side is right.

This work asks a complementary question at higher fidelity but lower N: within the specific structural family that DMI-candidate screening pipelines actually select – noncentrosymmetric space groups, a net magnetic moment, and a heavy spin-orbit-coupling element present – how often is MP’s stored FM label the true DFT ground state? We build a seven-source harmonized materials database and a bias-aware magnetic-ordering classifier, use them to filter for DMI-compatible candidates, and then run the same collinear energy-mapping protocol MP itself prescribes [6] on 24 of the resulting shortlisted candidates. The result is a direct, per-compound

DFT verdict rather than a statistical classifier disagreement, obtained specifically in the population most consequential for DMI-material discovery. We then carry one confirmed-FM candidate, $\text{Li}_4\text{Fe}_2\text{TeWO}_{12}$ (mp-754090), through a full pipeline: DFT extraction of both the isotropic exchange constant J and the DMI constant D , and finite-temperature atomistic spin dynamics driven by those DFT-derived constants with no fitting or rescaling.

II. METHODS

A. Harmonized multi-source database

Seven public sources – Materials Project (37,783 records, via `mp-api`), JARVIS-DFT (19,116), AFLOW (2,639), MAGNDATA (1,919 of 2,414 attempted), 2DMatPedia (1,862), OQMD (816), and NOMAD (500) – were merged into one schema (`material_id`, `formula`, `elements`, `space_group`, `density`, `volume`, `band_gap`, `ordering`, `total_magnetization`, and, where available, full structure and per-site magnetic-moment vectors), for 64,635 total records. Records retain a source tag rather than being silently deduplicated across sources. MAGNDATA is the only source with experimentally determined (rather than DFT-derived) magnetic structure, and is accordingly the ground-truth source for classifier training and validation. A new end-to-end scraper against the live Bilbao Crystallographic Server index recovered 1,919 of 2,414 entries (79.5%); the 495 failures are almost entirely (466 of 495) disordered or partial-occupancy structures that standard CIF parsing cannot represent, an upstream data-format limitation rather than a scraper defect. Ordering was validated on a known case (LaMnO_3 , correctly recovered as AFM, space group $Pnma$).

B. Structure-aware magnetic-ordering classifier

A LightGBM classifier predicts ordering (FM/AFM/FiM/NM) from composition- and structure-derived features. A composition-only baseline (per-element statistics of atomic number, mass, electronegativity, atomic radius, plus group/period/transition-metal/rare-earth fractions) achieves 56.7% accuracy (60.7% balanced accuracy). Adding two tiers of structural features – broad, near-universal-coverage fields (space group number, crystal-system bucket, unit-cell volume, site count) and narrow, MAGNDATA-only local-geometry features around magnetic sites (nearest magnetic-magnetic neighbor distance, coordination number, computed from atomic positions and species only, never from the magnetic-moment vectors that generate the label itself) – raises this to 67.0% accuracy (68.1% balanced accuracy) on a stratified 28,769/7,193 train/test split. Both feature tiers contribute independently, and space-group number becomes the single most important feature by LightGBM gain, despite

being absent from the baseline entirely. This classifier is retrained on all labeled data and deployed to predict ordering for every record lacking a ground-truth label, including all 19,116 JARVIS records, none of which carry a populated ordering field.

C. DMI-compatibility screening funnel

The full 64,635-record database, with ordering backfilled by the deployment classifier where no ground-truth label exists, is filtered for candidates plausibly compatible with a DMI: a net magnetic moment (FM or FiM; AFM is excluded, since a texture-hosting net moment is a hard requirement for the downstream simulation stage), a noncentrosymmetric space group (DMI is symmetry-forbidden under an inversion center [2]; verified computationally against all 230 space groups, recovering the textbook 92 centrosymmetric / 138 noncentrosymmetric split), presence of at least one heavy spin-orbit-coupling element (Ta, W, Re, Os, Ir, Pt, Au, Hg, Bi – the classic DMI-enhancing species by the Fert–Levy and Moriya mechanisms [2, 10]), and thermodynamic plausibility ($E_{\text{hull}} \leq 0.1$ eV/atom where known, 92.6% coverage). Of 64,635 records, 1,647 pass the full funnel. For the top 100 screened candidates per source, structures were backfilled via each source’s own public data-access method (299 of 300 attempted backfills succeeded) and re-relaxed with a MACE foundation-model potential [11], fine-tuned on this project’s harmonized dataset via multihead training (a replay head sampling 20,000 configurations from the original pretraining distribution to avoid catastrophic forgetting), as an independent consistency check against each source’s own DFT relaxation. All 298 relaxations converged (max force ≤ 0.05 eV/Å) with a median relaxation-energy shift of 0.0009 eV/atom relative to the source structure, indicating strong agreement between the fine-tuned potential and each source’s own DFT. Classifier confidence and MACE relaxation-consistency were merged into a final ranking (converged $\rightarrow$ hull-stable $\rightarrow$ smallest relaxation shift $\rightarrow$ classifier confidence); the resulting top-30 shortlist is the pool this work’s DFT validation batch is drawn from.

D. DFT ordering-validation protocol

For a subset of shortlisted candidates carrying an `ordering_source=ground_truth`, `confidence=1.0` label from Materials Project (i.e., MP’s own recorded ordering, not a classifier prediction), we apply the same collinear magnetic-ordering energy-mapping approach behind MP’s own workflow [6]: symmetry-distinct FM and AFM spin configurations are enumerated with `pymatgen.MagneticStructureEnumerator` (backed by `enumlib`); VASP inputs are generated with `MPRelaxSet` – MP’s own exchange-correlation functional, GGA+U parameters (e.g. LDAU= 5.3 eV on Fe), plane-wave cut-

off (520 eV), and k -point convention – overriding only the initial MAGMOM per configuration; and each configuration is relaxed to convergence. This choice keeps the resulting energies directly comparable to MP’s own stored values rather than constituting an arbitrary independent DFT recipe. Twenty-four candidates completed (85 individual VASP calculations, every one converged, verified against false electronic convergence by inspecting the RMS charge-density residual rather than trusting the convergence banner alone); two further candidates (mp-1173143, mp-682554) remain incomplete at the time of writing, with only their FM configuration converged, and are excluded from the analysis below.

All five tungsten-bearing candidates in this batch initially failed to converge under MP’s standard Hubbard correction (LDAU= 6.2 eV on W): tungsten here is formally W^{6+} , a nominally d^0 ion, and a large U applied to a near-empty d -shell leaves the self-consistent density with no restoring force, oscillating indefinitely rather than converging. Removing U from tungsten (LDAUL= -1) entirely, applied uniformly across every ordering of the affected compound, converged all 18 corresponding production calculations cleanly on the first retry. This changes the absolute energy scale for those five candidates relative to MP’s own U-inclusive value, but not the internal FM-vs-AFM comparison within each compound, since the same correction is applied to every ordering of that compound identically.

E. Exchange and DMI extraction for mp-754090

For $Li_4Fe_2TeWO_{12}$ (mp-754090), confirmed FM by the protocol above, we additionally extract effective spin-Hamiltonian parameters directly from DFT. The isotropic nearest-shell exchange constant is obtained from the same converged FM/AFM total energies used for the ordering verdict: the two Fe sites (Fe0, Fe1) per formula-unit cell have a $z = 4$ nearest opposite-sublattice coordination shell at 5.11–5.16 Å (mean 5.132 Å), cleanly separated from the next shell (starting 5.411 Å); for $H = -\sum_{ij} J'_{ij} \hat{s}_i \cdot \hat{s}_j$, the AFM-vs-FM energy difference per cell gives $J' = \Delta E/z = 12.06$ meV, treating the shell as one effective bond (a mean-field-style single- J fit, not a multi-shell decomposition).

DMI does not appear in a collinear FM-vs-AFM comparison at all; it requires spins canted relative to one another. We first checked whether mp-754090’s crystal symmetry forces $D = 0$ via Moriya’s rule (an inversion center at the bond midpoint is sufficient to forbid DMI exactly, at zero DFT cost) [2]: the specific Li/Fe/Te/W cation ordering in this structure reduces the space group to $P1$, with no symmetry operation beyond the identity, so no such shortcut applies and a real calculation is required. We then ran two static, constrained noncollinear-spin+spin-orbit-coupling calculations from the relaxed FM structure, using the penalty-energy method (I_CONSTRAINED_M=1, constraining mo-

ment direction only) to hold Fe0 along $+\hat{z}$ and Fe1 along $+\hat{x}$ (canted₊) or $-\hat{x}$ (canted₋) – i.e. $\theta = \pm 90^\circ$ canting in the xz -plane, an arbitrary but fixed reference axis since $P1$ symmetry does not privilege any particular plane. The moment constraint held essentially exactly in both calculations (Fe1 converged to $4.038 \mu_B$ along x in canted₊ and $-4.039 \mu_B$ in canted₋; Fe0 leakage off z was $< 0.002 \mu_B$ in both). For $E(\theta) = E_0 - zJ \cos \theta - zD \sin \theta$, comparing $\theta = \pm 90^\circ$ isolates the antisymmetric term cleanly since $\cos(\pm 90^\circ) = 0$:

$$zD = \frac{E(\theta = -90^\circ) - E(\theta = +90^\circ)}{2}, \quad (1)$$

giving $D = 0.193$ meV. Because $\hat{s}_1 = \hat{z}$ and the canting is confined to the xz -plane, $\hat{s}_1 \times \hat{s}_2 = (0, \sin \theta, 0)$: this probe measures the y -component of the DMI vector in this lab frame specifically, not a generic magnitude, and should be read as a resolved DFT number for one projection of D rather than a fully characterized three-dimensional DMI vector.

F. Finite-temperature spin dynamics

The DFT-derived $J = 12.06$ meV and $D = 0.193$ meV, with no fitting or rescaling, parameterize an exchange+DMI atomistic spin Hamiltonian for the Fe sublattice of mp-754090 (framework Li/Te/W/O sites carry no spin and are omitted from the geometry), implemented in LAMMPS-SPIN [12] via `spin/exchange` and `spin/dmi` pair styles on a $16 \times 8 \times 8$ replication (2048 spins) of the two-atom Fe primitive cell, with periodic boundaries and a rigid (spin-only) lattice. The system is initialized with random spin orientations (a paramagnetic start), relaxed at $T = 0$ with a spin-LBFGS minimizer, then evolved under Langevin spin dynamics (damping 0.05, $dt = 0.1$ fs) at ten temperatures from 10 to 400 K for 20,000 steps (2 ps) each, tracking the normalized net magnetization $|m|$.

III. RESULTS

A. DFT validation of the shortlisted FM population

Of the 24 completed candidates, all carrying an MP-recorded `ground_truth`, confidence-1.0 ferromagnetic label, 17 have their FM label overturned by a lower-energy antiferromagnetic configuration (margins 0.31–24.68 meV/atom), 4 are confirmed as the true FM ground state (1.18–2.41 meV/atom below the next-lowest AFM configuration), and 3 sit within 0.10 meV/atom – inside the EDIFF = 10^{-6} eV electronic-convergence tolerance used throughout – and are read as degenerate rather than as confirming either ordering (Fig. 1). Among the two remaining shortlisted-but-labeled-FM candidates (mp-22972, mp-551086), DFT confirms AFM is lower than

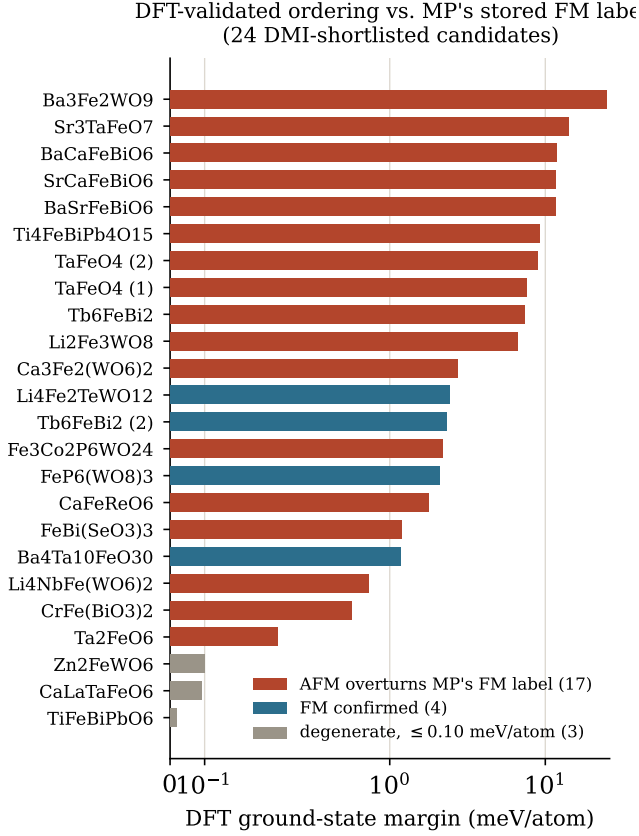


FIG. 1. DFT ground-state margin for the 24 completed candidates, sorted by magnitude (symmetric-log horizontal axis). Seventeen candidates' MP-recorded FM label is overturned by a lower-energy AFM configuration (red); four are confirmed as the true FM ground state (blue); three sit at or below the electronic-convergence tolerance (≤ 0.10 meV/atom, gray) and are read as degenerate rather than informative.

FM but does not test the recorded FiM label directly, since a collinear FM/AFM enumeration cannot represent a ferrimagnetic configuration; both are counted as overturned relative to the FM alternative only. A representative case, mp-861559 (TaFeO₄), illustrates the setup's internal consistency: its converged FM configuration reproduces MP's own recorded total magnetization to five significant figures (29.99998 vs. 30.0022 μ_B), ruling out a convention mismatch, while three symmetry-distinct AFM configurations all converge below it, the lowest by 0.274 eV/cell (≈ 45.7 meV/Fe).

Among the 21 candidates that resolve clearly (excluding the 3 degenerate ties), overturned outcomes outnumber confirmed FM outcomes 17 to 4. This sample is drawn from the top of the DMI-filtered shortlist (noncentrosymmetric, heavy-SOC-element, net-moment structures) rather than a random or stratified sample of the full `ground_truth` population, so it establishes the mislabeling rate within this specific structural family rather than a project-wide estimate; it is plausible, though

TABLE I. Summary of DFT-validated ordering outcomes for the 24 completed candidates.

Outcome	Count
AFM overturns MP's recorded FM/FiM label	17
FM confirmed as DFT ground state	4
Degenerate (≤ 0.10 meV/atom)	3
Total completed	24
Incomplete at time of writing	2

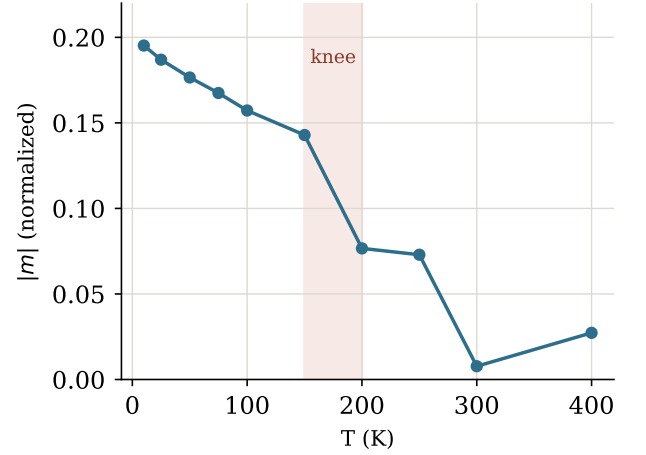


FIG. 2. Normalized net magnetization of the mp-754090 Fe sublattice (2048-spin cell) versus temperature, driven purely by DFT-derived $J = 12.06$ meV and $D = 0.193$ meV with no fitting. $|m|$ decays monotonically with a knee between 150 and 200 K. Absolute magnitude is suppressed throughout by multi-domain cancellation from the random-spin initial condition (Sec. IV C); the curve's shape, not its scale, is the physically meaningful signal here.

untested here, that this family is more AFM-prone than the ground-truth population as a whole. What it establishes unambiguously is that MP's FM label cannot be taken at face value for candidates drawn from exactly this population, which is precisely the population that DMI/skyrmion-material screening pipelines – including the one used to generate this shortlist – select from by construction.

B. mp-754090: DFT-derived exchange, DMI, and finite-temperature ordering

For mp-754090, confirmed FM by 2.41 meV/atom, the DFT-derived exchange and DMI constants are $J = 12.06$ meV and $D = 0.193$ meV, giving $D/J \approx 1.6\%$. This is notably smaller than an earlier order-of-magnitude placeholder estimate used before this calculation ($D/J \sim 25\%$), and is physically plausible: the magnetic ion at the probed bond, Fe, is a comparatively light 3d species with weak intrinsic spin-orbit coupling, despite the com-

TABLE II. Normalized net magnetization $|m|$ of the mp-754090 Fe sublattice vs. temperature, 2 ps Langevin dynamics per point, DFT-derived J and D .

T (K)	$ m $	T (K)	$ m $
10	0.1952	150	0.1429
25	0.1869	200	0.0766
50	0.1765	250	0.0729
75	0.1675	300	0.0078
100	0.1572	400	0.0273

pound containing tungsten, a heavy-SOC element that is precisely what qualified this candidate for the DMI-compatibility funnel in the first place. This is a useful cautionary note for the screening heuristic itself: the presence of a heavy-SOC element anywhere in the composition does not guarantee a large DMI constant at the specific magnetic-ion bond that anchors the spin texture.

Table II and Fig. 2 show the resulting finite-temperature magnetization sweep. $|m|$ decreases monotonically from 0.195 at 10 K to 0.008 at 300 K, with the sharpest drop – roughly a factor of two – occurring between 150 and 200 K, consistent with an ordering transition in that range. The uptick at 400 K (0.027, above the 300 K value) is not read as physical; Sec. IVC discusses why.

IV. DISCUSSION

A. Two independent routes to the same conclusion

Fahmy [7] detects the Materials Project’s ferromagnetic bias statistically: a classifier trained on experimentally solved magnetic structures (MAGNDATA) disagrees with MP’s stored label for thousands of compounds, at large N but without a first-principles check on any individual case. The present work detects the same qualitative phenomenon by the opposite route – a small- N , high-fidelity DFT energy-mapping check, applied specifically within the noncentrosymmetric, heavy-SOC-element, net-moment structural family that DMI/skymion-material discovery pipelines target. That these two very different methodologies, applied to different (though overlapping-in-spirit) populations, converge on the same conclusion – MP’s single-initial-guess relaxation workflow systematically under-counts antiferromagnetic ground states – is stronger evidence than either result alone. It also sharpens the practical implication specifically for DMI-material discovery: the population this work checked is not a random cross-section of MP but exactly the slice that such pipelines filter down to, and a 17-of-24 overturn rate in that slice is a direct warning to any workflow, including our own, that uses MP’s ordering label as a hard screening criterion rather than a prior.

B. Practical implication for DMI-candidate screening

The screening funnel used to generate this shortlist (Sec. IIC) excludes AFM-labeled candidates from consideration entirely, on the reasoning that a net moment is required for the texture-hosting mechanisms this project targets. If 17 of 24 top-ranked, MP-labeled-FM candidates from that funnel are, in fact, AFM, a nontrivial fraction of the true magnetic-material landscape relevant to this search is being filtered out before any DFT or simulation resource is spent on it – and, conversely, downstream compute (structure backfill, MLIP relaxation, spin-dynamics simulation) risks being spent on candidates whose net-moment premise is simply wrong. The corrective is not to trust the classifier over MP either – both are, in the end, estimates – but to treat any **ordering** field, ground-truth-labeled or classifier-predicted, as a prior to be checked with real DFT before committing expensive downstream stages, exactly as done here for mp-754090.

C. Limitations

The 24-candidate validation batch is drawn from the top of one DMI-filtered shortlist, not a random or stratified sample of the broader **ground_truth** population (1,647 candidates pass the DMI funnel; 24 of those, not 24 of the full database, were checked). The mislabeling rate reported here is a property of this structural family, not a project-wide estimate. Two further candidates (mp-1173143, mp-682554) remain incomplete and are excluded rather than assumed either way.

The mp-754090 DMI probe resolves one projection (D_y in an arbitrary lab frame, fixed consistently across the two canted calculations) of what is, in $P1$ symmetry, an unconstrained three-component DMI vector; a complete characterization would require probing additional canting planes. Single-ion anisotropy was not probed and is set to zero in the spin dynamics, for lack of any DFT or literature basis for a value.

The magnetization sweep’s absolute scale is suppressed throughout, and its non-monotonic uptick at 400 K is not physical: relaxing a large (2048-spin), fully periodic cell from a random-spin initial condition with no symmetry-breaking field commonly produces a multi-domain final state whose domains partially cancel in the net magnetization, well below the true per-domain order parameter. The knee between 150 and 200 K is therefore read as a qualitative signal of an ordering transition in that range, not a quantitative critical temperature; a reliable T_c estimate would require either an applied symmetry-breaking seed field, multi-seed ensemble averaging, or finite-size-scaling analysis (e.g. Binder-cumulant crossing), none of which were performed here.

V. CONCLUSION

Within the specific structural family that Dzyaloshinskii–Moriya-candidate screening pipelines select – noncentrosymmetric, net-moment, heavy-spin-orbit-coupling-element compounds – direct DFT energy-mapping on 24 Materials-Project-labeled ferromagnets finds 17 mislabeled (true ground state antiferromagnetic), 4 confirmed, and 3 degenerate. This DFT-level result independently corroborates, in a smaller but more directly consequential population, the large-scale statistical finding of systematic ferromagnetic bias in MP recently reported from ML classifiers trained on experimentally solved magnetic structures [7]. For one confirmed candidate, mp-754090, we additionally

demonstrate a complete DFT-to-dynamics pipeline: exchange and DMI constants extracted from first principles, with no fitting, drive a finite-temperature spin-dynamics simulation that reproduces a physically sensible, monotonically decaying magnetization curve with an ordering knee near 150–200 K. Both results argue for treating high-throughput database ordering labels as a screening prior rather than ground truth, particularly within the structural families that downstream physical mechanisms (here, DMI) specifically select for.

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- [1] I. Dzyaloshinsky, A thermodynamic theory of “weak” ferromagnetism of antiferromagnetics, *J. Phys. Chem. Solids* **4**, 241 (1958).
 - [2] T. Moriya, Anisotropic superexchange interaction and weak ferromagnetism, *Phys. Rev.* **120**, 91 (1960).
 - [3] A. Fert, V. Cros, and J. Sampaio, Skyrmions on the track, *Nat. Nanotechnol.* **8**, 152 (2013).
 - [4] N. Nagaosa and Y. Tokura, Topological properties and dynamics of magnetic skyrmions, *Nat. Nanotechnol.* **8**, 899 (2013).
 - [5] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, *APL Mater.* **1**, 011002 (2013).
 - [6] M. K. Horton, J. H. Montoya, M. Liu, and K. A. Persson, High-throughput prediction of the ground-state collinear magnetic order of inorganic materials using Density Functional Theory, *npj Comput. Mater.* **5**, 64 (2019).
 - [7] A. E. Fahmy, Learning Magnetic Order Classification from Large-Scale Materials Databases, arXiv:2509.05909 (2026).
 - [8] S. V. Gallego, J. M. Perez-Mato, L. Elcoro, E. S. Tasci, R. M. Hanson, K. Momma, and M. I. Aroyo, MAGNDATA: towards a database of magnetic structures. I. The commensurate case, *J. Appl. Crystallogr.* **49**, 1750 (2016).
 - [9] S. V. Gallego, J. M. Perez-Mato, L. Elcoro, E. S. Tasci, R. M. Hanson, and G. Madariaga, MAGNDATA: towards a database of magnetic structures. II. The incommensurate case, *J. Appl. Crystallogr.* **49**, 1941 (2016).
 - [10] A. Fert and P. M. Levy, Role of anisotropic exchange interactions in determining the properties of spin-glasses, *Phys. Rev. Lett.* **44**, 1538 (1980).
 - [11] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi, MACE: Higher order equivariant message passing neural networks for fast and accurate force fields, *Adv. Neural Inf. Process. Syst.* **35** (2022).
 - [12] J. Tranchida, S. J. Plimpton, P. Thibaudau, and A. P. Thompson, Massively parallel symplectic algorithm for coupled magnetic spin dynamics and molecular dynamics, *J. Comput. Phys.* **372**, 406 (2018).